

Haochuan (Harry) Zhang

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EDUCATION

Oregon State University

Master of Engineering in Computer Science, GPA: 3.8

Corvallis, OR

Sep 2024 – Jun 2026 (expected)

University of Kansas

Bachelor of Science in Computer Science

Lawrence, KS

Aug 2017 – May 2022

TECHNICAL SKILLS

Programming: Java, Python, C/C++, SQL, JavaScript, TypeScript, Haskell, Matlab

Frameworks: Spring Boot, JPA, Hibernate, Angular, PyTorch, Node.js

Tools: Docker, Redis, Kafka, Jenkins, Terraform, Git, Microservices

Cloud: AWS (S3, EC2, Lambda)

Databases: MySQL, PostgreSQL, MongoDB

Parallel & HPC: OpenMP, SIMD, CUDA, MPI

ML/AI: CNN, Transformers, GCCA, YOLO, PyTorch

PROFESSIONAL SUMMARY

Master's student in Computer Science with experience in backend software development and machine learning research. Skilled in Python, Java, and C++ with practical knowledge of PyTorch, CUDA, and distributed systems. Research interests include representation learning, computer vision, and embodied AI. Passionate about building systems that bridge engineering with intelligent, interactive applications.

EXPERIENCE

Research Intern – Brain-Computer Interface Project

Chongqing University of Posts and Telecommunications, China

Dec 2023 – Jun 2024

- Developed a neural control interface using BCI for stroke patients, improving processing speed by 9% and accuracy by 12%.
- Implemented neural decoding in C and Python, reducing misclassification by 5%.
- Optimized ML pipelines for feature extraction and user response, improving interaction by 15%.
- Contributed to two patents on EEG signal processing.

Software Developer

Charter Spectrum, Denver, CO

Aug 2022 – Jul 2023

- Refactored legacy SOAP APIs to RESTful services using Spring Boot, reducing ramp-up time by 50%.
- Led microservices migration, improving scalability and reducing technical debt.
- Implemented CI test pipelines with JUnit and Mockito, accelerating release cycles.
- Collaborated with frontend to decouple backend services and cut UI load time by 50%.

PROJECTS

- **Multiview (Alt)MAX-VAR GCCA Reproduction:** Reproduced and analyzed multiview representation learning methods (GCCA, MaxVar-GCCA, AltMaxVar-GCCA) using TensorFlow. Aligned multimodal embeddings (image, audio, text) via matrix factorization; visualized results and benchmarked alignment quality. Explored relevance for AI perception and multimodal understanding.
- **Semantic Segmentation Platform (UNet):** Built a PyTorch-based UNet training pipeline for semantic segmentation tasks; implemented custom data loaders, training/validation split, and performance metrics. Evaluated the model on small-scale visual datasets.
- **TastyBytes Web Application:** Designed a full-stack recipe sharing web app with Angular and Node.js; enabled users to upload photos, videos, and comments with AWS S3 backend. Enhanced user interaction through responsive UI.

SELECTED GRADUATE COURSEWORK

CS 575: Parallel Programming (A), CS 539: Nonlinear Optimization (A), AI 586: Applied Matrix Analysis (A-), CS 325: Analysis of Algorithms (A), CS 261: Data Structures (A), CS 271: Computer Architecture (A)

PATENTS

Contributor: CN117743818A – Domain Adaptation Method for Motor Imagery EEG Signals (2024)